THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL
FORM TWO PRE-MOCK EXAMINATION 1
032 CHEMISTRY

Time: 2 Hours April 2026

Instructions:

1. This paper consists of sections A and B with a total of ten questions.
2. Answer ALL questions in the spaces provided.
3. All writing must be in blue or black ink, except drawing, which must be in pencil.
4. Cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your Examination Number at the top right corner of every page.
6. The following atomic masses may be used: H =1, C = 12, N = 14, O = 16, S =32, Ca = 40, Na = 23, Cl = 35.5, Mg = 24, Al = 27, K = 39, Cu = 64, Fe = 56

SECTION A (15 MARKS)

1. Choose the most correct answer among the given alternative and write down the answer in the space given.

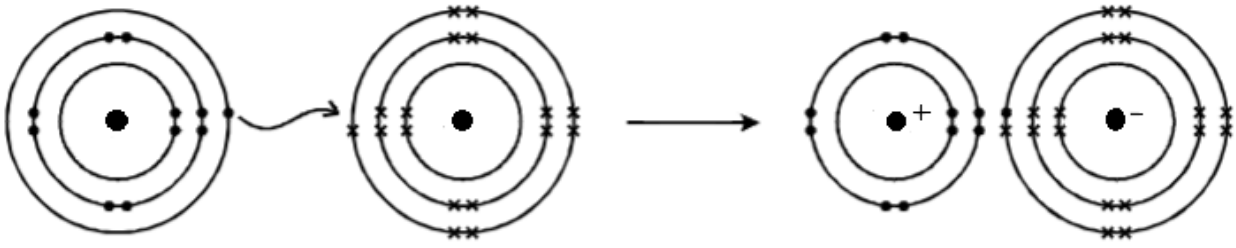
(i) Which of the following substances when heated changes directly into vapor without becoming liquids first?

A. Iodine and ice
B. Ammonium chloride and candle wax
C. Dry ice and sodium chloride
D. Ammonium chloride and iodine

(ii) Which one of the following is not a solution

A. Bronze B. Brass C. Zinc D. Air

(iii) The diagram below shows atoms bonding to form a compound. Which of the following is correct about the formed compound?



A. Sodium chloride, ionic compound
B. Sodium chloride, covalent compound
C. Potassium chloride, ionic compound
D. Potassium chloride, covalent compound

(iv) This chemical can stimulate a burning substance to burn efficiently and faster:
   A. Flammable substance, for example petrol
   B. Oxidizing agent, for example oxygen
   C. Explosive substance, for example sodium metal
   D. Highly flammable substance, for example hydrogen gas

(v) The most abundant element on earth is:
   A. Hydrogen
   B. Sodium
   C. Nitrogen
   D. Oxygen

(vi) A chemistry student at Seypalm secondary school accidentally splashed an alkaline solution into her eyes. Which is the most immediate safety action to take?
   A. Neutralize it by splashing an acidic solution into the eyes.
   B. Rub the eyes vigorously with a clean dry towel.
   C. Wash the eyes with plenty of running water using an eye wash station.
   D. Cover the eyes with a bandage and go home.

(vii) Why is the hole of a Bunsen Burner closed when the burner is not in use for heating?
   A. To produce a very hot flame.
   B. To save gas by reducing the oxygen supply.
   C. To produce a luminous flame which is visible to prevent accidents.
   D. To prevent the gas from leaking into the laboratory.

(viii) Which set of elements represents a single group in the Periodic Table?
   A. Lithium, Sodium, Potassium
   B. Carbon, Nitrogen, Oxygen
   C. Helium, Neon, Chlorine
   D. Magnesium, Aluminium, Silicon

(ix) A mixture of Ethanol (boiling point 78°C) and Water (boiling point 100°C) is best separated by:
   A. Simple distillation
   B. Fractional distillation
   C. Using a separating funnel
   D. Evaporation to dryness

(x) Which of the following is a balanced equation for the reaction between Sodium and Water?
   A. $Na+H_2O \rightarrow NaOH+H_2$
   B. $2Na+2H_2O \rightarrow 2NaOH+H_2$
   C. $Na+2H_2O \rightarrow Na(OH)_2+H_2$
   D. $2Na+H_2O \rightarrow Na_2O+H_2$

2. Match the items in list A with the correct response in list B by writing the letter of the correct response beside the item number in answer booklet provided.

List A	List B
(i) A solution which gives white precipitate with dilute hydrochloric acid (ii) A solution which gives a blue precipitate with sodium hydroxide solution (iii) A compound which decomposes to form an oxide which is yellow when hot and white when cold (iv) A compound which forms a black oxide on heating (v) A solution which forms a pale green precipitate with ammonia solution	A. Lead acetate B. Copper (II) chloride C. Iron (III) chloride D. Zinc carbonate E. Copper (II) carbonate F. Iron (II) chloride G. Ethane H. Ethene

SECTION B (70 MARKS)

3. (a) Define "Chemistry" and state its contribution to the manufacturing industry in Tanzania.
   (b) Give three reasons why studying chemistry is essential for a student aiming for a career in environmental conservation.
4. (a) State two differences between a "Physical Change" and a "Chemical Change."
   (b) Balance the following equations and write the complete ionic and net ionic equations for the each:
   (i) $K_2C_2O_{4(aq)} + Ba(OH)_{2(aq)} \rightarrow KOH_{(aq)} + BaC_2O_{2(s)}$
   (ii) $Pb(NO_3)_{2(aq)} + H_2SO_{4(aq)} \rightarrow PbSO_{4(s)} + HNO_{3(aq)}$
   (iii) $CaCO_{3(s)} + H_2SO_{4(aq)} \rightarrow CaSO_{4(s)} + CO_{2(g)} + H_2O_{(l)}$

5. (a) Briefly discuss two amendments could be made on Dalton's atomic theory, based on new findings due to the development of methods for more detailed investigation of the nature of matter.
   (b) (i) What is an "Isotope"?
   ……
   (ii) Chlorine has two isotopes: $Cl{-}35$ (75% abundance) and $Cl{-}37$ (25% abundance). Calculate the Relative Atomic Mass (RAM) of Chlorine.

6. (a) (i) What is an "Acid"?
   ……
   (ii) Complete the following equations:
   a) $NaOH_{(aq)} + HCl_{(aq)} \rightarrow$ ……
   b) $2HCl_{(aq)} + CaCO_{3(s)} \rightarrow$ ……
   c) $H_2SO_{4(aq)} + Ca(HCO_3)_{2(aq)} \rightarrow$ ……

   (b) A solution turns red litmus paper blue.
   (i) Is the solution acidic or basic? ……
   (ii) What would be the color of this solution if phenolphthalein indicator is added?

7. (a) Name and explain the three components of the "Fire Triangle" that are required for a fire to continue or start to burn.

   (b) Explain why water should never be used to extinguish a fire involving a localized oil spill in a kitchen.

8. Write the letters given after the table with the information to fit the elements in the table use the information given in each part of the question.

Table:

I							**VIII**
	II	III	IV	V	VI	VII	

(a) Write;
   (i) **A** in the table for an element found in group (II) and period 3.
   (ii) **B** for an element found in group (III) and period 2.
   (iii)**C** for an element found in group (VII) and period 2.
   (iv)**D** for an element found in group (I) and period 3.
   (v) **E** for an element found in group (VII) and period 3.
   (vi) Write the actual symbols of element A-E

   A……B……C…….……...D……E……

   (vii) The tendency of showing a regular repeating pattern of elements in the table above is called…….

(b) Briefly explain the importance of using chemical symbols, Give giving four points.

9. (a) In Longido secondaery school chemistry laboratory, you are given a mixture containing ammonium chloride and maize flour and you are required to obtain the ammonium chloride for laboratory use, free of the corn flour. Name and describe the quickest and most suitable mixture separation method you would use to obtain pure ammonium chloride.

   (b) Giving three points, explain the importance of changing matter from one state to another.

SECTION C (15 MARKS)

10. Before starting to fight a fire, it is important to know the size and class of the fire that you are going to put off. Describe those fire classes.